Linear Independent

The Definition of Linear Independence

⚠ **Definition.** A set of vectors $\{v_1, v_2, \ldots, v_k\}$ is ***linearly independent*** if the vector equation

$$x_1v_1 + x_2v_2 + \cdots + x_kv_k = 0$$

has only the trivial solution $x_1 = x_2 = \cdots = x_k = 0$. The set $\{v_1, v_2, \ldots, v_k\}$ is ***linearly dependent*** otherwise.

In other words, $\{v_1, v_2, \ldots, v_k\}$ is linearly dependent if there exist numbers $x_1, x_2, \ldots, x_k$, not all equal to zero, such that

$$x_1v_1 + x_2v_2 + \cdots + x_kv_k = 0.$$

This is called a ***linear dependence relation*** or ***equation of linear dependence***.

Theorem. *Let $v_1, v_2, \ldots, v_k$ be vectors in $\mathbf{R}^n$, and consider the matrix*

$$A = \begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_k \\ | & | & & | \end{pmatrix}.$$

Then we can delete the columns of A without pivots (the columns corresponding to the free variables), without changing $\mathrm{Span}\{v_1, v_2, \ldots, v_k\}$.

The pivot columns are linearly independent, so we cannot delete any more columns without changing the span.

Let $A = \begin{bmatrix} | & | & & | \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \\ | & | & & | \end{bmatrix}$

$\{\vec{v}_1, \cdots, \vec{v}_n\}$ is L.I.

⇕

pivot in every column of A

⇕

$A\vec{x} = \vec{0}$ has only the trivial solution

$\{\vec{v}_1, \cdots, \vec{v}_n\}$ is L.D.

⇕

free variables exist

⇕

$A\vec{x} = \vec{0}$ has infinitely many solutions

if $\{\vec{v}_1, \vec{v}_2, \cdots, \vec{v}_n\}$ is linearly independent

the $A = \begin{bmatrix} | & | & & | \\ \vec{v}_1 & \vec{v}_2 & \cdots & \vec{v}_n \\ | & | & & | \end{bmatrix}$ has pivot in every column.

Subspaces

Definition. A ***subspace*** of $\mathbf{R}^n$ is a subset V of $\mathbf{R}^n$ satisfying:

1. ***Non-emptiness:*** The zero vector is in V.
2. ***Closure under addition:*** If u and v are in V, then $u+v$ is also in V.
3. ***Closure under scalar multiplication:*** If v is in V and c is in $\mathbf{R}$, then cv is also in V.

Common Types of Subspaces

Theorem (Spans are Subspaces and Subspaces are Spans). *If $v_1, v_2, \ldots, v_p$ are any vectors in $\mathbf{R}^n$, then $\text{Span}\{v_1, v_2, \ldots, v_p\}$ is a subspace of $\mathbf{R}^n$. Moreover, any subspace of $\mathbf{R}^n$ can be written as a span of a set of p linearly independent vectors in $\mathbf{R}^n$ for $p \leq n$.*

Definition. Let A be an $m \times n$ matrix.

- The ***column space*** of A is the subspace of $\mathbf{R}^m$ spanned by the columns of A. It is written $\text{Col}(A)$.
- The ***null space*** of A is the subspace of $\mathbf{R}^n$ consisting of all solutions of the homogeneous equation $Ax = 0$:

$$\text{Nul}(A) = \{x \text{ in } \mathbf{R}^n \mid Ax = 0\}.$$

Basis & Dimension

⚠ **Definition.** Let V be a subspace of $\mathbf{R}^n$. A ***basis*** of V is a set of vectors $\{v_1, v_2, \ldots, v_m\}$ in V such that:

1. $V = \text{Span}\{v_1, v_2, \ldots, v_m\}$, and
2. the set $\{v_1, v_2, \ldots, v_m\}$ is linearly independent.

⚠ **Definition.** Let V be a subspace of $\mathbf{R}^n$. The number of vectors in any basis of V is called the ***dimension*** of V, and is written $\dim V$.

Let A be an $m \times n$ matrix with r pivots.

	$\text{Col } A$	$\text{Nul } A$
Definition	All linear combinations of the columns of A	All solutions to $Ax = \mathbf{0}$
Subspace of	$\mathbb{R}^m$	$\mathbb{R}^n$
Dimension	$\text{rank}(A) = r$ = number of pivots	$\text{nullity}(A) = n - r$ = number of free variables
How to find a basis	Find the pivot positions by row reducing A; select the corresponding columns of the original A	Solve $Ax = \mathbf{0}$ in parametric vector form; take the vector multiplying each free variable

$$\boxed{\dim(\text{Col } A) + \dim(\text{Nul } A) = n}$$

Basis Theorem. *Let V be a subspace of dimension m. Then:*

- *Any m linearly independent vectors in V form a basis for V.*
- *Any m vectors that span V form a basis for V.*

The Rank Theorem

Definition. The ***rank*** of a matrix A, written $\text{rank}(A)$, is the dimension of the column space $\text{Col}(A)$.

The ***nullity*** of a matrix A, written $\text{nullity}(A)$, is the dimension of the null space $\text{Nul}(A)$.

Rank Theorem. *If A is a matrix with n columns, then*

$$\text{rank}(A) + \text{nullity}(A) = n.$$

In other words, for any consistent system of linear equations,

$$(\text{dim of column span}) + (\text{dim of solution set}) = (\text{number of variables}).$$

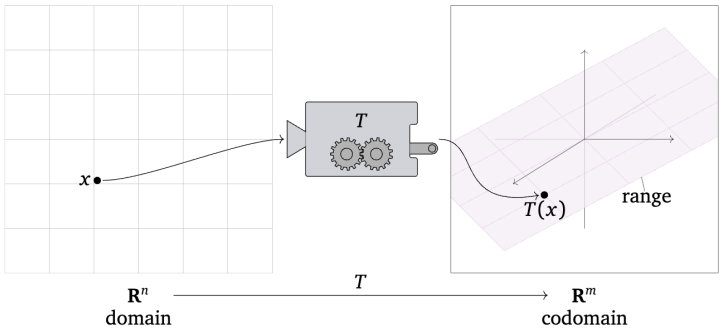
Matrix Transformations

Definition. A ***transformation*** from $\mathbf{R}^n$ to $\mathbf{R}^m$ is a rule T that assigns to each vector x in $\mathbf{R}^n$ a vector $T(x)$ in $\mathbf{R}^m$.

- $\mathbf{R}^n$ is called the ***domain*** of T.
- $\mathbf{R}^m$ is called the ***codomain*** of T.
- For x in $\mathbf{R}^n$, the vector $T(x)$ in $\mathbf{R}^m$ is the ***image*** of x under T.
- The set of all images $\{T(x) \mid x \text{ in } \mathbf{R}^n\}$ is the ***range*** of T.

The notation $T : \mathbf{R}^n \longrightarrow \mathbf{R}^m$ means "T is a transformation from $\mathbf{R}^n$ to $\mathbf{R}^m$."

It may help to think of T as a "machine" that takes x as an input, and gives you $T(x)$ as the output.



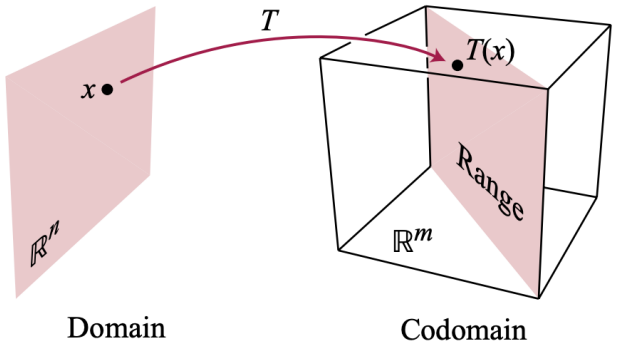
Definition. Let A be an $m \times n$ matrix. The ***matrix transformation*** associated to A is the transformation

$$T : \mathbf{R}^n \longrightarrow \mathbf{R}^m \quad \text{defined by} \quad T(x) = Ax.$$

This is the transformation that takes a vector x in $\mathbf{R}^n$ to the vector Ax in $\mathbf{R}^m$.

Let A be an $m \times n$ matrix, and let $T(x) = Ax$ be the associated matrix transformation.

- The *domain* of T is $\mathbf{R}^n$, where n is the number of *columns* of A.
- The *codomain* of T is $\mathbf{R}^m$, where m is the number of *rows* of A.
- The *range* of T is the *column space* of A.



One-to-one and Onto Transformation

Definition (One-to-one transformations). A transformation $T : \mathbf{R}^n \to \mathbf{R}^m$ is ***one-to-one*** if, for every vector b in $\mathbf{R}^m$, the equation $T(x) = b$ has *at most one* solution x in $\mathbf{R}^n$.

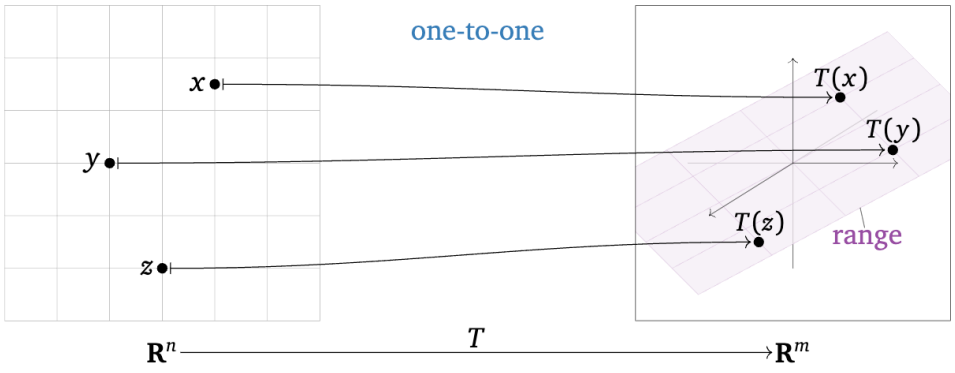
Definition (Onto transformations). A transformation $T : \mathbf{R}^n \to \mathbf{R}^m$ is ***onto*** if, for every vector b in $\mathbf{R}^m$, the equation $T(x) = b$ has *at least one* solution x in $\mathbf{R}^n$.

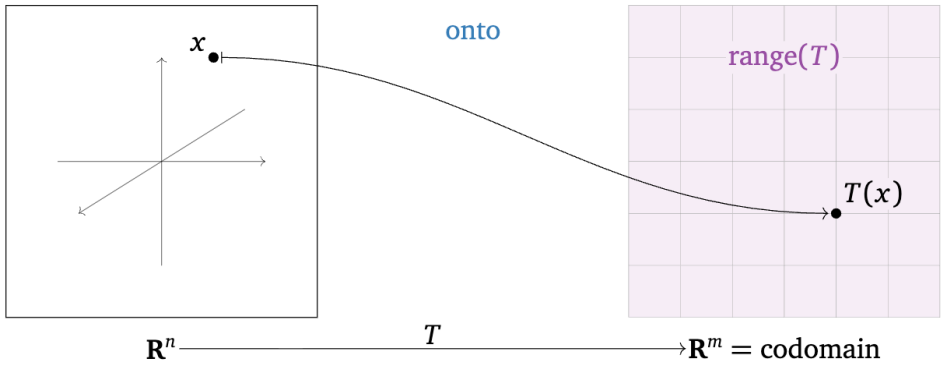
Theorem (One-to-one matrix transformations). *Let A be an $m \times n$ matrix, and let $T(x) = Ax$ be the associated matrix transformation. The following statements are equivalent:*

1. *T is one-to-one.*
2. *For every b in $\mathbf{R}^m$, the equation $T(x) = b$ has at most one solution.*
3. *For every b in $\mathbf{R}^m$, the equation $Ax = b$ has a unique solution or is inconsistent.*
4. *$Ax = 0$ has only the trivial solution.*
5. *The columns of A are linearly independent.*
6. *A has a pivot in every column.*
7. *The range of T has dimension n.*

Theorem (Onto matrix transformations). *Let A be an $m \times n$ matrix, and let $T(x) = Ax$ be the associated matrix transformation. The following statements are equivalent:*

1. *T is onto.*
2. *$T(x) = b$ has at least one solution for every b in $\mathbf{R}^m$.*
3. *$Ax = b$ is consistent for every b in $\mathbf{R}^m$.*
4. *The columns of A span $\mathbf{R}^m$.*
5. *A has a pivot in every row.*
6. *The range of T has dimension m.*





Linear Transformation

Definition. A ***linear transformation*** is a transformation $T : \mathbf{R}^n \to \mathbf{R}^m$ satisfying

$$T(u+v) = T(u) + T(v)$$
$$T(cu) = cT(u)$$

for all vectors u, v in $\mathbf{R}^n$ and all scalars c.

Facts about linear transformations. *Let $T : \mathbf{R}^n \to \mathbf{R}^m$ be a linear transformation. Then:*

1. $T(0) = 0$.
2. *For any vectors $v_1, v_2, \ldots, v_k$ in $\mathbf{R}^n$ and scalars $c_1, c_2, \ldots, c_k$, we have*

$$T(c_1v_1 + c_2v_2 + \cdots + c_kv_k) = c_1T(v_1) + c_2T(v_2) + \cdots + c_kT(v_k).$$

Standard coordinate vectors.

The ***standard coordinate vectors in*** $\mathbf{R}^n$ are the n vectors

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \\ 0 \end{pmatrix}, \quad \ldots, \quad e_{n-1} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \end{pmatrix}, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

The ith entry of e_i is equal to 1, and the other entries are zero.

From now on, for the rest of the book, we will use the symbols $e_1, e_2, \ldots$ to denote the standard coordinate vectors.

Multiplying a matrix by the standard coordinate vectors. *If A is an $m \times n$ matrix with columns $v_1, v_2, \ldots, v_m$, then $Ae_i = v_i$ for each $i = 1, 2, \ldots, n$:*

$$\begin{pmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & & | \end{pmatrix} e_i = v_i.$$

In other words, multiplying a matrix by e_i simply selects its ith column.

If $A = [\vec{a}_1 \ \vec{a}_2 \ \vec{a}_3 \ \cdots \ \vec{a}_n]$

$T(\vec{e}_1) = \vec{a}_1, \ \cdots, \ T(\vec{e}_n) = \vec{a}_n$

We saw in the above example that the matrix for counterclockwise rotation of the plane by an angle of θ is

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

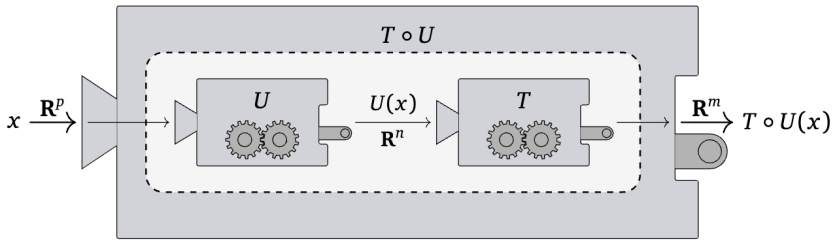
Matrix Multiplication

Composition of linear transformations

Composition means the same thing in linear algebra as it does in Calculus. Here is the definition.

Definition. Let $T: \mathbf{R}^n \to \mathbf{R}^m$ and $U: \mathbf{R}^p \to \mathbf{R}^n$ be transformations. Their ***composition*** is the transformation $T \circ U: \mathbf{R}^p \to \mathbf{R}^m$ defined by

$$(T \circ U)(x) = T(U(x)).$$



Domain and codomain of a composition.

- In order for $T \circ U$ to be defined, the codomain of U must equal the domain of T.
- The domain of $T \circ U$ is the domain of U.
- The codomain of $T \circ U$ is the codomain of T.

e.g. $T: \mathbb{R}^n \to \mathbb{R}^m$, $U: \mathbb{R}^p \to \mathbb{R}^n$, $T \circ U: \mathbb{R}^p \to \mathbb{R}^m$

Properties of composition. Let S, T, U be transformations and let c be a scalar. Suppose that $T: \mathbf{R}^n \to \mathbf{R}^m$, and that in each of the following identities, the domains and the codomains are compatible when necessary for the composition to be defined. The following properties are easily verified:

$$S \circ (T + U) = S \circ T + S \circ U \qquad (S + T) \circ U = S \circ U + T \circ U$$
$$c(T \circ U) = (cT) \circ U \qquad c(T \circ U) = T \circ (cU) \quad \text{if } T \text{ is linear}$$
$$T \circ \mathrm{Id}_{\mathbf{R}^n} = T \qquad \mathrm{Id}_{\mathbf{R}^m} \circ T = T$$
$$S \circ (T \circ U) = (S \circ T) \circ U$$

Composition of transformations is *not* commutative in general. That is, in general, $T \circ U \neq U \circ T$, even when both compositions are defined.

The sizes of the matrices in the matrix product.

- In order for AB to be defined, the number of rows of B has to equal the number of columns of A.
- The product of an $m \times n$ matrix and an $n \times p$ matrix is an $m \times p$ matrix.

Recipe: The row-column rule for matrix multiplication. Let A be an $m \times n$ matrix, let B be an $n \times p$ matrix, and let $C = AB$. Then the ij entry of C is the ith row of A times the jth column of B:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

Here is a diagram:

$$\begin{pmatrix} a_{11} & \cdots & a_{1k} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ik} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mk} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{1j} & \cdots & b_{1p} \\ \vdots & & \vdots & & \vdots \\ b_{k1} & \cdots & b_{kj} & \cdots & b_{kp} \\ \vdots & & \vdots & & \vdots \\ b_{n1} & \cdots & b_{nj} & \cdots & b_{np} \end{pmatrix} = \begin{pmatrix} c_{11} & \cdots & c_{1j} & \cdots & c_{1p} \\ \vdots & & \vdots & & \vdots \\ c_{i1} & \cdots & c_{ij} & \cdots & c_{ip} \\ \vdots & & \vdots & & \vdots \\ c_{m1} & \cdots & c_{mj} & \cdots & c_{mp} \end{pmatrix}$$

ith row; jth column; ij entry

Matrix multiplication caveats.

- Matrix multiplication is not commutative: AB is not usually equal to BA, even when both products are defined and have the same size. See this example.
- Matrix multiplication does not satisfy the cancellation law: $AB = AC$ does not imply $B = C$, even when $A \neq 0$. For example,

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 5 & 6 \end{pmatrix}.$$

- It is possible for $AB = 0$, even when $A \neq 0$ and $B \neq 0$. For example,

$$\begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

Products and compositions.

The matrix of the composition of two linear transformations is the product of the matrices of the transformations.

$T(x) = Ax \quad U(x) = Bx$

$T \circ U(x) = T(U(x)) = T(Bx) = ABx$